

ECON 0150 | Spring 2026 | Homework 4.3

Due: Sunday April 5, at 11:59 PM

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely. You are welcomed and encouraged to work in groups so long as your work is your own. Use the provided datasets to answer the following questions. Then submit your figures and answers to Gradescope.

Q1. Happiness and GDP

This question is based on our work during Exercise 4.1 and Exercise 4.3 in class. Use the happiness dataset `Life_Evaluation_GDP_PerCap.csv` to answer the following questions. The dataset contains the following variables:

- Year: year of observation
- Country: the country of observation
- Life_Evaluation: composite measure of happiness based on survey polling
- GDP_Per_Cap: the country's GDP in that year divided by the population

a) Create a scatterplot with GDP Per Capita on the horizontal axis and happiness on the vertical axis for the year 2024.

b) Does there appear to be a relationship between the two variables?

c) Does the relationship appear to be linear?

d) Perform a log transformation of GDP Per Capita and create a scatterplot with log GDP Per Capita on the horizontal axis and happiness on the vertical axis.

e) Fit a linear regression model to predict happiness using log GDP Per Capita. What is the interpretation of the intercept coefficient?

f) What is the interpretation of the slope coefficient?

g) What is the interpretation of the p-value of the slope coefficient?

Q2. Model Diagnostics

Using the model you fit in Q1(e), answer the following questions about model assumptions.

a) Extract the residuals and predicted (fitted) values from your model. Create a residual plot with predicted values on the x-axis and residuals on the y-axis. Add a horizontal line at zero.

b) Does the residual plot show signs of heteroskedasticity? Explain what you see and which assumption this relates to.

c) Create a histogram of the residuals. Do they appear roughly normal?

d) Create a lagged residual plot by plotting each residual against the previous residual. Does this model appear to violate the independence assumption?

e) Re-fit the model using robust standard errors by adding `cov_type='HC3'` to the `.fit()` call. How do the standard errors and p-values change compared to Q2(e)?

Q3. Reading a Residual Plot

A researcher fits a linear model predicting test scores from hours of study. Their residual plot (predicted values on the x-axis, residuals on the y-axis) shows a clear fan shape: residuals are tightly clustered near zero for low predicted values but spread out widely for high predicted values.

a) Which model assumption does this residual plot suggest is violated?

b) What is the name for this pattern?

c) Does this violation affect the estimated coefficients ($\hat{\beta}_0$ and $\hat{\beta}_1$), or does it affect the standard errors and p-values? Explain briefly.

d) What is one approach the researcher could use to address this issue?